StratAI

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What is the project about?

We will have lot of fun learning Game Theory, with a focus on concepts such as Nash Equilibrium and optimal strategies. We will further learn and apply the fundamentals of Machine Learning to develop a prediction model. Train and test the model on unseen games to test its ability. We will also have a sneak peek into Reinforcement Learning and apply it to create a model that learns strategies in repeated games, where players adapt based on the outcomes of previous rounds.

Game theory and Nash equilibrium

As the name suggests Game theory is studying how to win games using well defined mathematical strategies rather than just brute guesswork!

Formally, it is the study of mathematical models of strategic interactions, where we have a set of decision-makers (players) who change the **game state** and the reward associated with every player might be affected by the decision others take. So the challenge is not just doing your best but doing best given the possible (best) actions of other players !!

Note that we assume that all players are perfectly rational and will always act optimally.
It has wide applications in economics and finance, and understanding decision making in humans and animals.

(Caution: historical content ahead)

Modern game theory began with the idea of mixed-strategy equilibria in two-person zero-sum games and its proof by John von Neumann. He used the Brouwer's fixed point theorem on continuous mappings into compact convex sets, which is now a standard method in game theory and mathematical economics.

Often it turns out that players can benefit by mutual cooperation but due to lack of trust opt a **dominant strategy** which always stands best irrespective of the decisions of other players.

When all the players act according to the dominant strategy the game arrives at a **Nash Equilibrium**.

Task 1 : Read about Nash equilibrium and prisoner's dilemma here.

Task 2 : Now for the 2 player pay off tables below, try to determine dominant strategies for each player, and the set of decisions that corresponds to a Nash Equilibrium.

1. Atreya and Prad run their own whiskey shops. Assume that they are the only sellers and their profits depend upon the prices they set. The pair (a, p) means a is Atreya's and p is Prad's profit.

Observe that the profit increases when price is decreased. This happens in practical situations also, because decreased price increases demand which generally outweighs the effect of reduce in price (~~Those interested should take HS3002~~)

Atreya's dominant: 500
Prad's dominant: 1000
NASH: (500, 500)

Atreya's Price \ Prad's Price	500	1000
500	$(10k, 10k)$	$(12k, 5k)$
1000	$(5k, 12k)$	$(8k, 8k)$

2. Consider the game of rock, paper and scissors, where the scores of two players depend upon the pair of choices they take. Scores are represented as (Player 1's score, Player 2's score)

Player 1 \ Player 2	rock	paper	scissors
rock	$(0, 0)$	$(0, 1)$	$(1, 0)$
paper	$(1, 0)$	$(0, 0)$	$(0, 1)$
scissors	$(0, 1)$	$(1, 0)$	$(0, 0)$

Does a dominant strategy always exist? NO

Is Nash always optimal?

A tournament

Lets imagine you are going to play the Prisoner's Dilemma with your friend not once but multiple times. Your score at the end of the game is the total number of years in jail. You would like to make this score as small as possible. What strategy would you play?

In a game repeated many times, a strategy that allows players to return to the cooperative outcome after a period of non-cooperation may be preferable.

To see what strategies work best, political scientist Robert Axelrod held a tournament. People entered by sending computer programs designed to play repeated prisoners' dilemma. Each program then played the game against all the other programs. The "winner" was the program that received the fewest total years in jail.

The winner turned out to be a simple strategy called **tit-for-tat**. According to tit for tat, a player should start by cooperating and then do whatever the other player did last time. Thus, a tit for tat player cooperates until the other player defects; then she defects until the other player cooperates again. In other words, this strategy starts out friendly, penalizes unfriendly players, and forgives them if warranted.

To Axelrod's surprise, this simple strategy did better than all the more complicated strategies that people had sent in.

The tit for tat strategy has a long history. It is essentially the biblical strategy of **"an eye for an eye, a tooth for a tooth."** The prisoners' dilemma tournament suggests that this may be a good rule of thumb for playing some of the **games of life.** :)

Task 1 : So is Nash in fact always optimal? No

Task 2 : What is the optimal strategy for Atreya and Prad when they run whiskey shops (in the Task 2 of the previous section)? (500, 500)

Mixed Strategies

Matching pennies

Consider Pranjal and Shivanshu play a game. Each of them have a coin, and depending on what the 2 voluntarily show, they get points based on the following table.

Pranjal \ Shivanshu	Heads	Tails
Heads	$(1, -1)$	$(-1, 1)$
Tails	$(-1, 1)$	$(1, -1)$

Suppose Pranjal believed Shivanshu plays Heads with probability p. Then if Pranjal plays Heads, Pranjal gets 1 with probability p and –1 with probability (1–p), for an expected value of 2p–1. Similarly, if Pranjal plays Tails, he gets –1 with probability p (when Shivanshu plays Heads), and 1 with probability (1–p), for an expected value of 1–2p.

If 2p–1 > 1–2p, then Pranjal is better off, on average, playing Heads than Tails. Similarly, if 2p–1 < 1–2p, then he is better off playing Tails than Heads. If, on the other hand, 2p–1 = 1–2p, then he gets the same payoff no matter what he does. In this case, he could play Heads, could play Tails, or could flip a coin and randomize his play.

A **mixed strategy Nash equilibrium** involves at least one player playing a randomized strategy and no player being able to increase his or her expected payoff by playing an alternate strategy. A Nash equilibrium in which no player randomizes is called a **pure strategy Nash equilibrium.**

Pranjal \ Shivanshu	Heads	Tails	
Heads	$(1, -1)$	$(-1, 1)$	$1(p) + (-1)(1-p) = 2p - 1$
Tails	$(-1, 1)$	$(1, -1)$	$-1(p) + 1(1-p) = 1 - 2p$

Note that randomization requires equality of expected payoffs. If a player is supposed to randomize over strategy A or strategy B, then both of these strategies must produce the same expected payoff. Otherwise, the player would prefer one of them and wouldn't play the other.

Note that it is Shivanshu's probabilities that determine the expected payoffs for Pranjal; if Pranjal is going to randomize, then Shivanshu's probabilities must be such that Pranjal is willing to randomize.

For Pranjal to randomize (payoffs must be equal) p should be $\frac{1}{2}$.

Battle of sexes

In IITM, there lives a happy couple and on the same day 2 events clash : A basketball game and a ballet.

Boy \ Girl	Basketball	Ballet
Basketball	(3,2)	(1,1)
Ballet	(0,0)	(2,3)

Let the girl go to the basketball game with probability p and the boy with probability q.

Boy \ Girl	Basketball(p)	Ballet($1-p$)	Boy's Payoff
Basketball(q)	(3,2)	(1, 1)	$3p + 1(1-p) = 1 + 2p$
Ballet($1-q$)	(0,0)	(2, 3)	$0(p) + 2(1-p) = 2 - 2p$
Girl's Payoff	$2(q) + 0(1-q) = 2q$	$1(q) + 3(1-q) = 3 - 2q$	

(Verify the calculations of payoffs above.)

A mixed strategy requires both parties to randomize (since a pure strategy by either party prevents randomization by the other).

The boy's indifference between going to the Basketball game and to the Ballet requires $1+2p = 2-2p$, which yields p = $\frac{1}{4}$. That is, the boy will be willing to randomize which event he attends if the girl is going to the Ballet $\frac{3}{4}$ of the time. This makes the boy indifferent between the two events because he prefers to be with the girl, but also likes to be at the Basketball game.

For the girl to randomize, q should be $\frac{3}{4}$. Thus, the probability that the boy goes to the Basketball game is $\frac{3}{4}$, and he goes to the Ballet $\frac{1}{4}$ of the time.

These are independent probabilities, so to get the probability that both go to the Basketball game, we multiply the probabilities, which yields $\frac{3}{16}$.

Boy \ Girl	Basketball	Ballet
Basketball	3/16	9/16
Ballet	1/16	3/16

Note that more than half of the time (Basketball, Ballet) is the outcome of the mixed strategy and the two people are not together. This lack of coordination is generally a feature of mixed strategy equilibria. Thus, both do worse than coordinating on their less preferred outcome. But this mixed strategy Nash equilibrium, undesirable as it may seem, is a Nash equilibrium in the sense that neither party can improve his or her own payoff, given the behavior of the other party.

(**Note : A simple call on the telephone should rule out the mixed strategy. :)**)

Task 1 : Find equilibria for the following games.

Pure Nash: (Swerve, Don't), (Don't, Don't)

Mixed Nash:

You play Swerve with prob p & Me play Swerve with prob q:

You's expected pay-off = $0 \cdot q + (-1)(1-q) = q - 1$
$= 1 \cdot q + (-4)(1-q) = 5q - 4$ $\Rightarrow q = 3/4$

me 's expected payoff = $0 \cdot p + (-1)(1-p) = p - 1 \Rightarrow p = 3/4$
$= 1 \cdot p + (-4)(1-p) = 5p - 4$

pure Nash: (Swerve, Swerve), (Don't, Don't)

You \ Me	Swerve	Don't
Swerve	$(0, 0)$	$(-1, 1)$
Don't	$(1, -1)$	$(-4, -4)$

You \ Me	Swerve	Don't
Swerve	$(3, 3)$	$(0, 0)$
Don't	$(4, 5)$	$(8, 0)$

Task 2 : If you multiply a player's payoff by a positive constant, does the equilibria of the game change? Why / Why not? (Consider the cases for both pure and mixed Nash equilibria)

pure: unchanged, cause players' preference over outcomes are not changed.

Now have some fun watching this !!

mixed: unchanged, as scaling just cancels out in the indifferent equations.